Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Quantitative software developer experienced in **Python data pipelines, ETL, C++, SQL, data validation, testing, CI/CD, and monitoring**, building reliable infrastructure for large-scale processing, enrichment, and efficient retrieval.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built Python ETL-style pipelines processing 5,000+ monthly jobs with incremental loads, validation, reconciliation, and error handling, increasing concurrent throughput 3x while reliably sustaining 99%+ reliability in production.
- Optimized PostgreSQL and SQL Server retrieval for 500+ users through indexing, caching, and query tuning, reducing dashboard latency from 1.2 seconds to 350 milliseconds and integration latency 35%.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented concurrent Python ingestion for 1,000+ PDF, DOCX, ZIP, text, and image files per batch, adding validation, fault-tolerant execution, diagnostics, and automated tests for reliable peak-load processing.
- Monitored OCR, model, timeout, and API failures across 10,000+ daily requests, implementing fallback paths and usage tracking while reliably reducing feedback turnaround time 60% at production scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built Python and PyTorch pipelines processing 100,000+ trajectory records, reliably improving long-horizon planning 18% through feature preparation, simulation, debugging, reproducible testing, and iterative model optimization at scale.
- Prepared dataset features from trajectory logs and HD-map vectors, improving closed-loop stability 15% while clearly documenting experimental tradeoffs, edge-case findings, and data-driven recommendations for senior researchers at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Redesigned Python ETL workflows using AWS Glue and S3 to ingest, transform, and clean JSON, CSV, and Parquet datasets, increasing pipeline efficiency 68% and reducing storage costs 15%.
- Contributed tested Python infrastructure with Jenkins, PyTest, and CI/CD across Linux, Windows, and macOS, improving API response times 30% while generating coverage summaries, failure logs, and nightly reports.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Engineered a Python feature-engineering pipeline over 650,000 URLs, designing 21 statistical indicators for IP presence, suspicious keywords, and structural anomalies to clean, tag, and classify high-volume inputs.
- Trained Random Forest and neural models, achieving 96%+ detection accuracy while comparing performance tradeoffs, validating results, and documenting reproducible findings from high-volume malicious and benign datasets at scale.

Skills

Languages: Python, C++, SQL, Java, Shell, Bash, Linux
Data/ML: ETL, data validation, reconciliation, feature engineering, PyTorch, Random Forest, statistical analysis
Storage: PostgreSQL, SQL Server, MongoDB, S3, Glue, incremental loads, efficient retrieval, data pipelines
Engineering: Git, CI/CD, Jenkins, PyTest, unit testing, integration testing, monitoring, performance optimization

Projects

CPU-scheduling Simulator Live | *C++, Performance Engineering*
- Implemented six C++ scheduling algorithms with edge-case handling, automated waiting and turnaround metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

Big-Brain Live | *Data Ingestion and Retrieval*
- Engineered a cloud ingestion and indexing pipeline over 50,000+ documents using PostgreSQL, GCP, LLaMA-2, FAISS, and Pinecone, achieving sub-second search while reducing retrieval latency 40% through performance tuning.

FinNexus Live | *Financial Data Platform*
- Built a fintech platform supporting 50,000+ users and one million transactions using TypeScript, MongoDB, Plaid, real-time updates, secure token controls, and Kubernetes-based CI/CD for reliable high-volume data delivery.